Discussion of "Some Insights on Flow over Sharp-Crested Weirs Using Computational Fluid Dynamics: Implications for Enhanced Flow Measurement"

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This paper presents a discussion of "Some Insights on Flow over Sharp-Crested Weirs Using Computational Fluid Dynamics: Implications for Enhanced Flow Measurement" by Joseph M. Sinclair, S. Karan Venayagamoorthy, and Timothy K. Gates. https://doi.org/10.1061/(ASCE)IR.1943-4774.0001652

Introduction

The discussers would like to thank and appreciate the authors for their efforts in finding the valuable and interesting results on flows over sharp-crested weirs using computational fluid dynamics. For weir flow (Bagheri and Heidarpour 2010), the hydraulic characteristics are well known and can be categorized into four flow patterns, including sharp-crested weir flow, spillway flow, broad-crested weir flow, and open channel flow as shown in Fig. 1. The State Key Laboratory of Hydraulics and Mountain River Engineering (Higher Education Press 2016) mentioned that sharp-crested weir flow occurs if the values of W_C/P (W_C is the crest width of a weir in the direction of flow and P is the weir height) are smaller than a critical value.

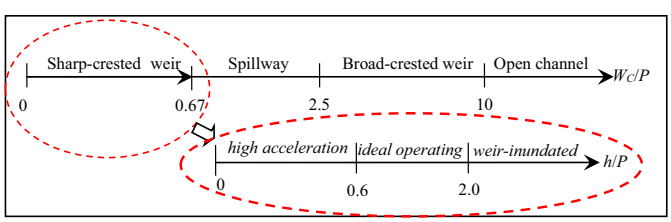

Fig. 1. Classification of the flow regimes related to sharp-crest weir.

According to the original paper, the authors investigated the hydraulic behavior and found that hydraulic behavior (velocity, pressure profiles, and discharge coefficients) can be classified as belonging to one of three different flow regimes, depending on h/P values (h is the elevation head over the weir): high-acceleration ($0 < h/P \leq 0.6$), ideal-operating ($0.6 < h/P \leq 2.0$), and weir-inundated ($h/P > 2.0$) flow regimes (Fig. 1). In addition, the authors analyzed three estimated discharge coefficients [C_d, C_{dR}, and C_{dT} using Eqs. (1)–(3)] and highlighted the need for caution when employing empirical discharge coefficients over a wide range of h/P values. They further discovered that the high-acceleration regime frequently resembled a clinging flow for some of the lower h/P values, and in the ideal-operating regime, using a value of $C_d = 0.75$ might be a good approximation. In fact, the discharge coefficient, C_{dR}, which is crucial for any practical measurements, is derived empirically (Zhang et al. 2010; Aydin et al. 2002).

Discharge coefficients are crucial and the key issues in providing accurate flow regimes over the weirs. The discussers consider that several formulas of the discharge coefficients used in the original paper lack application ranges. The value of C_d in the ideal-operating regime also suffers limitations and, consequently, the results of flow regimes and discharge coefficients in the original paper need to be discussed.

Recently, the discussers also investigated similar discharge coefficient problems, but over an inclinable sharp-crested weir (Fig. 1), thus providing additional data to analyze the authors' recommendations of discharge coefficients in different h/P values by varying the height of the weir (P). Regarding the numerical model for sharp-crested weirs discussed in this discussion, the onset of transition flow regime employs nearly identical characteristics as those considered in the original paper. The two-dimensional weir models were installed in a rectangular flume (8.0 m long and 1.0 m wide). The weir height was $P_0 = 0.3$ m and W_C was 0.02 m thick as shown in Fig. 2. The elevation head over the weir crest, h, was varied between 0.05 and 0.25 m. The oblique angle (θ) of the weir models varied from 0° to 90°, resulting in P values of 0.02–0.3 m. A total of 63 tests were conducted in the main test program by systematically varying the parameters relative to water head h/P and flow rate Q. The final configuration of the computational grid included a total number of grid nodes $N_{\text{total}} = 50{,}167$ for traditional sharp-crested weir ($\theta = 90°$) and $N_{\text{total}} = 49{,}358$ for inclinable sharp-crested weir ($\theta = 60°$) (Fig. 3).

The weir height P_0 and oblique angle (θ) were then used to calculate the values of relative crest width W_C/P in weir flow (Figs. 1 and 2). The calculated values indicate that there are a total of 16 tests of the flow patterns that match the sharp-crested weir flow. Table 1 presents the characteristic parameters used in numerical simulations, which only covered the sharp-crested weir flows, ranging from the smallest h/P values possible to the values for ideal-operating flow regime defined in the original paper. Three parameters (C_d, C_{dR}, and C_{dT}) were also calculated in these tests.

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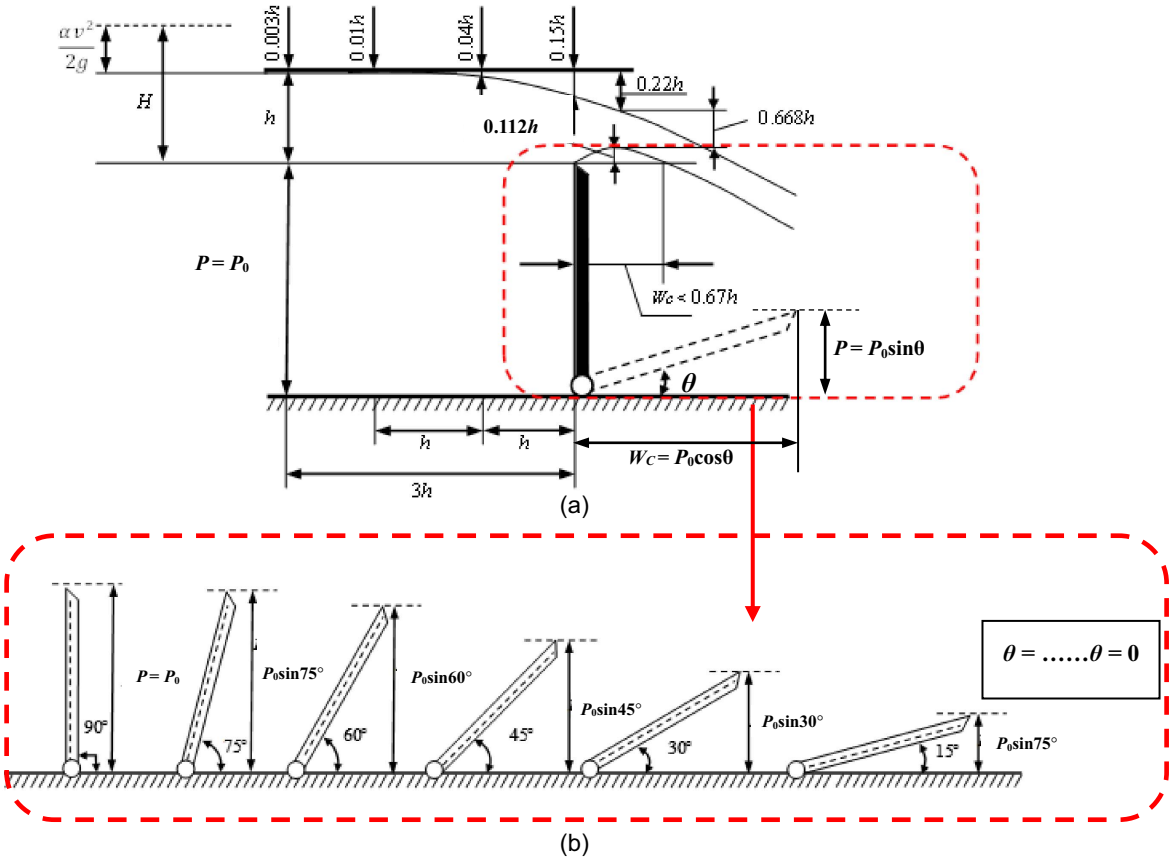

Fig. 2. Flows over a sharp-crested weir: (a) bottom axis–driven sharp-crested weir; and (b) different sharp-crested weir models (i.e., different P or θ).

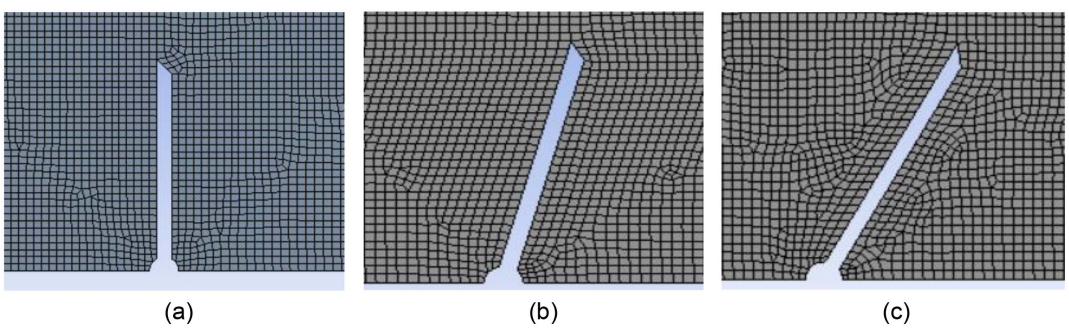

Fig. 3. Typical computational mesh: (a) traditional sharp-crested weir ($\theta = 90°$); (b) inclinable sharp-crested weir ($\theta = 75°$); and (c) inclinable sharp-crested weir ($\theta = 60°$).

Head-Discharge Behavior

In addition to verification, the various sets ($P = 0.2$ m and $h = 0.02$–0.22 m) of calculated results were first compared with the experimental results of Arvanaghi and Navid (2013) in terms of head-discharge relationships. Furthermore, the numerical sets in Table 1 were validated against a physical model: semiempirical formulation and empirical formulations. These included depths, velocities, and accurate representations of the head-discharge relationship in a sharp-crested weir.

The results in Fig. 4 indicate that the discrepancy between simulation data from the discussers and the data from the Arvanaghi and Navid (2013) is within 2%, indicating that the simulation process is effective and can correctly depict the real-world state of the weir flow of the models in Table 1.

Fig. 5 shows the relative water head h/P as a function of flow rate Q for oblique angle θ between 0° and 90°. Two characteristics are observed: (1) the flow rate Q increases with the relative water head h/P, and (2) the flow rate Q increases with increasing oblique angle θ, while practically no sharp-crested weir flow occurred when given $\theta < 60°$.

Discharge Formulas and Discharge Coefficients

The flow regimes (as a function of h/P for C_d) are compared in Fig. 6 using data from the authors, the discussers, and Arvanaghi and Navid. Based on Fig. 6, similar flow regimes are observed for the data of the discussers and the data of the authors, respectively (with negligible differences between the data of Arvanaghi and

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Table 1. Test program and parameter variation

θ (degrees)	P (m)	W_C (m)	Q (m^3/s)	H/P	h/P	A$u^2/2$ g (m)	C_d	C_{dR}	C_{dT}	Flow regime
60	0.260	0.150	0.239	1.0057	0.9625	0.011	0.6475	0.6832	0.6062	Ideal operating
75	0.290	0.078	0.091	0.4570	0.4487	0.002	0.6575	0.6447	0.6397	High acceleration
			0.106	0.5280	0.5177	0.003	0.6179	0.6498	0.6000	High acceleration
			0.127	0.6002	0.5868	0.004	0.6136	0.6550	0.5931	High acceleration
			0.167	0.7108	0.6903	0.006	0.6323	0.6628	0.6052	Ideal operating
			0.193	0.7846	0.7593	0.007	0.6334	0.6680	0.6031	Ideal operating
			0.240	0.8977	0.8629	0.010	0.6502	0.6757	0.6127	Ideal operating
90	0.300	0.002	0.023	0.1674	0.1667	0.000	0.6966	0.6235	0.6921	High acceleration
			0.045	0.2691	0.2667	0.001	0.6735	0.6310	0.6645	High acceleration
			0.060	0.3372	0.3333	0.001	0.6425	0.6360	0.6316	High acceleration
			0.088	0.4405	0.4333	0.002	0.6358	0.6435	0.6204	High acceleration
			0.109	0.5100	0.5000	0.003	0.6354	0.6485	0.6168	High acceleration
			0.131	0.5799	0.5667	0.004	0.6329	0.6535	0.6114	High acceleration
			0.167	0.6856	0.6667	0.006	0.6323	0.6610	0.6062	Ideal operating
			0.193	0.7568	0.7333	0.007	0.6334	0.6660	0.6042	Ideal operating
			0.234	0.8641	0.8333	0.009	0.6339	0.6735	0.6004	Ideal operating

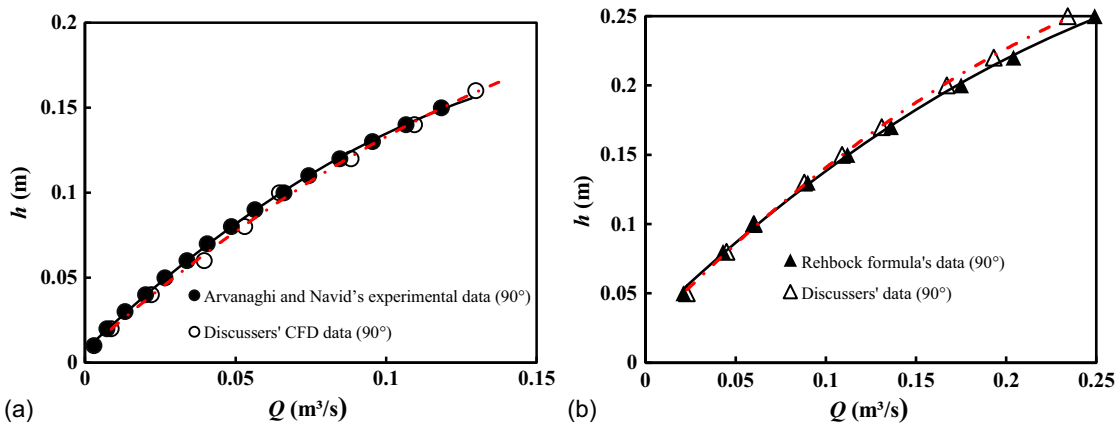

Fig. 4. Numerical model validation: (a) experimental data from Arvanaghi and Navid (2013) versus discussers' data; and (b) empirical formulation from Rehbock versus discussers' data.

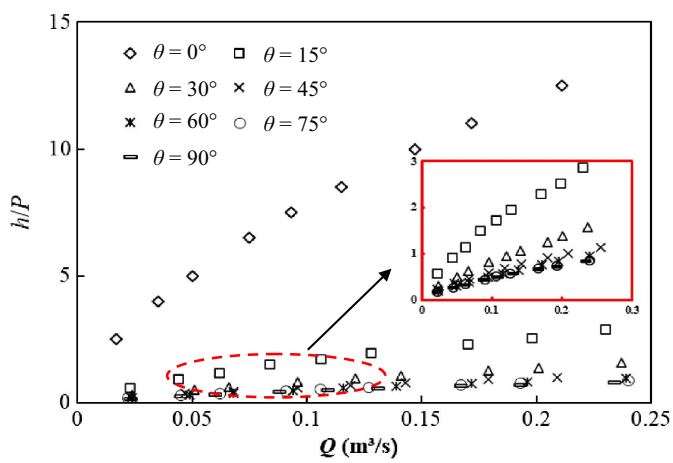

Fig. 5. Numerical model validation.

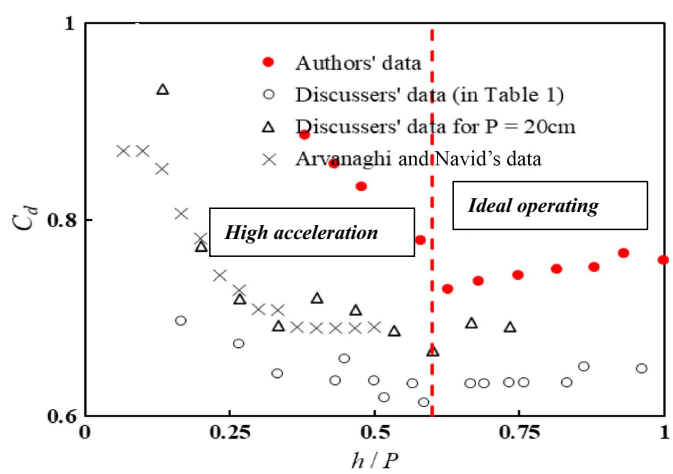

Fig. 6. Discharge coefficients as function of h/P for C_d.

Navid), where in general the values of discharge coefficient C_d decrease as the low-head h or h/P increase. In the ideal-operating regime, the C_d trend lines become essentially level $C_d = 0.75$, 0.65, 0.69, and 0.70 for the authors' data, the discussers' data in Table 1, the discussers' data for $P = 20$ cm, and Arvanaghi and Navid's data, respectively. This is because the surface tension and viscous related are more prevalent at low heads.

Therefore, the authors' conclusion in original paper about "using a value of $C_d = 0.75$ might be a good approximation" cannot be confirmed here. The results demonstrate that a comparison of

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Table 2. Classical discharge-coefficient formulas and the head limits for sharp-crested weir

Name	Date	Formula	Head limit (m)	Source
Bazin	1886	$C_{dR} = \frac{3}{2}\left(0.405 + \frac{0.003}{h}\right)\left(1 + 0.55\frac{h^2}{(h+P)^2}\right)$	$0.08 \le h \le 0.55$	Montes (1998)
SSAE	1926	$C_{dR} = \frac{3}{2}\left(0.410 + \frac{0.001}{h+0.0016}\right)\left(1 + 0.5\frac{h^2}{(h+P)^2}\right)$	$0.025 < h < 0.80$	Montes (1998)
Rejbock I	1920s	$C_{dR} = 0.605 + \frac{1}{1{,}000\ h} + 0.08\frac{h}{P}$	$0.025 < h < 0.60$	Finnemore and Franzini (2002)
Rejbock II	1920s	$C_{dR} = 0.611 + 0.08\frac{h}{P}$	$0.025 < h < 0.60$	Sturm (2001) Chaudhry (1993)
Henderson	1966	$C_{dR} = 0.611 + 0.075\frac{h}{P}$	$2 < h < 5$	Bos (1989)
Kindsvater and Carter	1957	$C_{DR} = \frac{3}{2\sqrt{2}} \times 0.554\left(1 - 0.003\frac{h}{P}\right)$	$h > 0.03$ $h \ge 0.03$ $h \ge 0.06$	Montes (1998) Herschy (1995) USBR (2001)

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Note: SSAE = Swiss Society of Architects and Engineers; and USBR = US Department of the Interior Bureau of Reclamation.

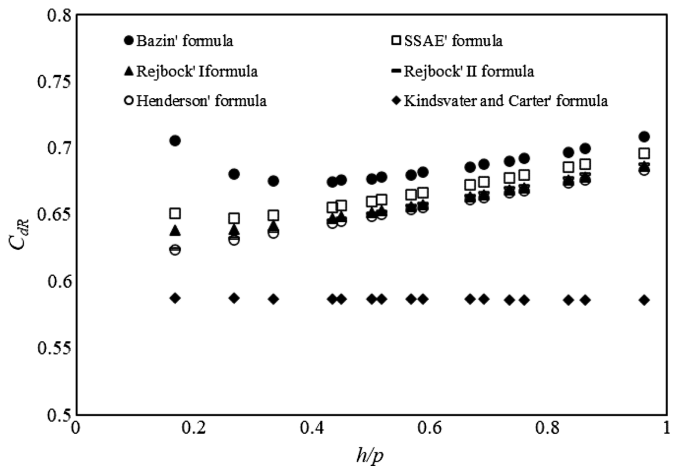

Fig. 7. Discharge coefficients as function of h/P for C_{dR} using the discussers' data in Table 1.

discharge coefficient for different methods can be varied as a function of the h or h/P.

There are numerous empirical formulas that can be used for analyzing the discharge coefficients in a sharp-crested weir [for example, C_{dR} and C_{dT} using Eq. (2) and (3) in the original paper]. Most of the equations were obtained by dimensional and statistical analysis of data acquired in physical models. Table 2 lists a few classical formulas for C_{dR} and the limitation on the head, h (Zhang et al. 2010; Aydin et al. 2002). If h is too low to form free flow, then clinging flow occurs, such that those C_{dR} formulas are not applicable.

Fig. 7 verifies that standard C_{dR} formulas have head limitations, as seen in Table 2. The reason for the trend variation is unclear; however, given that the range of limiting head values from Kindsvater and Carter (1957) is relatively small, this could be a result of fitting error.

Data Availability Statement

All data, solution procedures, and models used during the study appear in the published discussion or the original paper under discussion.

Acknowledgments

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